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**A Software Approach to the
PPT2 Conjecture**

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THE 2ND CYCLE MASTER'S STUDY PROGRAMME
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Abstract

Title: A Software Approach to the
PPT2 Conjecture

This thesis explores...

Keywords

PPT2 conjecture, quantum entanglement, positive maps, semidefinite programming, bound entanglement

Povzetek

Naslov: Programski pristop k domnevi PPT2

V tem delu raziskujemo...

Ključne besede

domneva PPT2, kvantna prepletenost, pozitivne preslikave, semidefinitno programiranje, vezana prepletenost

Razširjeni povzetek

Daljši slovenski povzetek vsebine...

1 TODO

Writing

- ☐ Write the abstracts: English, Slovene, and the extended Slovene summary.
- ☐ Polish the Introduction: spell out the implications of the conjecture holding versus failing, and sharpen the statement of current knowledge ($n \leq 3$ proven, $n \geq 4$ open).
- ☐ Expand the Conclusion into a genuine discussion: the Gram-representation freedom, the inertness of the symmetric (partial transpose invariant) family, the limitations, and the outlook.

Results (pending)

- ☐ Finalize the PPT2 search results: final candidates and witnesses, hardware, wall-clock, and outcome.
- ☐ Fill in the performance numbers in the Performance section: Julia generation timings and the MATLAB-prototype comparison.
- ☐ Remove the temporary `nbr()` red-highlight helper once all numbers are final.

References

- ☐ Complete and verify all citations: resolve the remaining reference placeholders and margin notes.

2 Introduction

Quantum entanglement is a central nonclassical resource in quantum information theory. It supports communication and cryptographic tasks and provides a structural lens for understanding quantum channels [1].

This thesis studies the PPT2 conjecture: whether the composition of two PPT maps is always entanglement breaking [2]. The conjecture has a clear operational interpretation, since entanglement-breaking maps destroy all bipartite entanglement with any reference system.

The stakes are structural. Were the conjecture to hold, composing two PPT maps would always destroy entanglement: the PPT cone, though strictly larger than the set of entanglement-breaking maps, would collapse into it after a single self-composition – a strong rigidity statement about how far a PPT map can sit from being entanglement breaking. Were it to fail, the counterexample would be a pair of PPT maps whose composition is itself PPT yet entangled, a bound entangled channel assembled from ordinary PPT ingredients, bearing on the structure of quantum channels and on the distillability questions that turn on the PPT property [1]. Either way the answer sharpens the picture of the boundary between positive and completely positive maps.

Known progress motivates a computation-first strategy. The conjecture is proven for $n \leq 3$ and for Choi-type maps in all dimensions [3], [4], but it remains open for $n \geq 4$, where the geometry of positive maps grows markedly more intricate and direct analytic classification has so far resisted. The smallest open case, 4×4 , is the natural target for a computational attack, and the one we pursue.

The core challenge is twofold. First, separability testing is hard in the regimes where the conjecture remains open. Second, numerical certificates near feasibility boundaries are fragile. Therefore, this thesis focuses on methodology: how to construct, compose, test, and validate candidate objects in a reproducible way.

The main objectives are:

1. Design a Julia workflow for candidate generation, map composition, and entanglement checks.
2. Integrate DPS-based semidefinite relaxations as a baseline witness route.

3. Integrate an SOS-based PNCP witness-construction route with post-solver validation.

The rest of the thesis is organized as follows. Chapter 2 develops the mathematical background: notation, matrix spaces, linear maps, entanglement, the criteria used to detect it, and the semidefinite-programming machinery, including the polynomial route to witnesses. Chapter 3 presents the computational methods and the implementation. Chapter 4 reports the experimental results. Chapter 5 discusses their significance, the limitations, and directions for future work.

3 Theoretical Background

This chapter establishes the mathematical framework underpinning the PPT2 conjecture and its computational study. We fix notation and define the key objects: linear maps, their positivity properties, quantum states, and entanglement criteria, then develop the semidefinite programming tools used in the implementation.

3.1 Notation

We write $\mathbb{C}^n$ and $\mathbb{R}^n$ for the complex and real coordinate spaces. $M_n(\mathbb{F})$ is the algebra of $n \times n$ matrices over the field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, abbreviated M_n when the field is clear from context. A linear map between matrix algebras is written $\Phi : M_n \rightarrow M_m$. We write $E_{ij} \in M_n$ for the matrix unit with a 1 in position (i, j) and zeros elsewhere.

In some parts of this work we also use *Dirac notation*, as it is the standard convention in quantum mechanics. With that in mind, let $|\psi\rangle$ represent the column vector $\psi \in \mathbb{C}^n$, and $\langle\psi|$ its adjoint row vector ψ^* . Now $\langle\psi | \varphi\rangle = \psi^* \varphi \in \mathbb{C}$ is the inner product and $|\psi\rangle\langle\psi| = \psi\psi^* \in M_n$ is the rank-one projector onto ψ ; in particular $E_{ij} = |i\rangle\langle j|$. A bipartite *product vector* is a simple tensor $|\psi\rangle \otimes |\varphi\rangle$, also written $|\psi \otimes \varphi\rangle$ or $|\psi\varphi\rangle$.

3.2 Matrix spaces

We equip M_n with involution $*$, that is, conjugate transposition ($A^* = \overline{A^T}$) over $\mathbb{C}$ and transposition ($A^* = A^T$) over $\mathbb{R}$. We denote the subspace of Hermitian matrices ($A = A^*$) by $H_n \subseteq M_n(\mathbb{C})$, and the subspaces of symmetric ($A = A^T$) and skew-symmetric ($A = -A^T$) matrices by $S_n, K_n \subseteq M_n(\mathbb{R})$ respectively, so that $M_n(\mathbb{R}) = S_n \oplus K_n$. Finally, $M_n^+ \subseteq H_n$ is the cone of positive semidefinite matrices.

Definition 3.1 (*Positive semidefinite and positive definite*) A matrix $A \in H_n$ is *positive semidefinite* (PSD), written $A \succeq 0$, if $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$, or equivalently if all eigenvalues of A are non-negative. It is *positive definite* (PD), written $A \succ 0$, if $x^*Ax > 0$ for all $x \in \mathbb{C}^n \setminus \{0\}$, equivalently if all eigenvalues are strictly positive.

The *Hilbert-Schmidt inner product* on M_n is defined as $\langle A, B \rangle := \text{tr}(B^*A)$, making M_n an inner product space. In the case of H_n or S_n , it simplifies to $\text{tr}(BA)$.

We work with bipartite systems on the tensor product vector space $\mathbb{C}^m \otimes \mathbb{C}^n$. We equip the operator space $M_m \otimes M_n \simeq M_{mn}$ with multiplication, i.e. $(A \otimes B)(C \otimes D) = AC \otimes BD$, for $A, C \in M_m$ and $B, D \in M_n$. We use the *partial trace* tr_k , a generalization of the trace operation, to *trace out* the k -th factor, for example $\text{tr}_1(A \otimes B) = \text{tr}(A)B$.

Definition 3.2 (*Partial transpose*) For a product matrix $A \otimes B \in M_m \otimes M_n$, the *partial transpose* with respect to subsystem B is defined as

$$(A \otimes B)^{\Gamma_B} := A \otimes B^T, \quad (1)$$

and extended by linearity to a general $\rho \in M_m \otimes M_n$.

For Hermitian ρ , the partial transpose with respect to either subsystem yields the same result: $\rho^{\Gamma_A} = \rho^{\Gamma_B}$, so we abbreviate to ρ^Γ without loss of generality.

3.3 Linear maps

Given linear maps $\Phi_1 : M_n \rightarrow M_m$ and $\Phi_2 : M_p \rightarrow M_q$, their *tensor product* $\Phi_1 \otimes \Phi_2 : M_n \otimes M_p \rightarrow M_m \otimes M_q$ is the linear map fixed on simple tensors by

$$(\Phi_1 \otimes \Phi_2)(A \otimes B) = \Phi_1(A) \otimes \Phi_2(B) \quad (2)$$

and extended by linearity. We write I_k for the identity map on M_k , and we say $I \otimes \Phi$ is the *ampliation* of Φ .

Definition 3.3 (*Positive, k -positive, and completely positive maps*) Let $\Phi : M_n \rightarrow M_m$ be a linear map.

- Φ is *positive* (P) if $A \succeq 0 \Rightarrow \Phi(A) \succeq 0$.
- Φ is *k -positive* if $I_k \otimes \Phi : M_k \otimes M_n \rightarrow M_k \otimes M_m$ is positive.
- Φ is *completely positive* (CP) if it is k -positive for every $k \in \mathbb{N}$ [3].

Clearly every completely positive map is also positive. However, there are many maps that are positive but not completely positive (PNCP). They will be playing a central role in this work.

Definition 3.4 (*Block positivity*) An operator $W \in H_{mn}$ is *block positive* if

$$\langle x \otimes y | W | x \otimes y \rangle \geq 0 \quad (3)$$

for all product vectors $x \in \mathbb{C}^m$ and $y \in \mathbb{C}^n$.

Definition 3.5 (*Choi-Jamiołkowski isomorphism*) Let $\Phi : M_n \rightarrow M_m$ be a linear map, then the ampliation of Φ on the maximally entangled state $|\Omega\rangle = \sum_{i=1}^n |i\rangle \otimes |i\rangle$ is its *Choi matrix*:

$$C_\Phi = (I \otimes \Phi)(|\Omega\rangle\langle\Omega|) = \sum_{i,j=1}^n |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|) \in M_n \otimes M_m. \quad (4)$$

The assignment $\Phi \mapsto C_\Phi$ is a linear isomorphism between maps $\Phi : M_n \rightarrow M_m$ and matrices in $M_n \otimes M_m$. The map can be recovered from its Choi matrix by $\Phi(\rho) = \text{tr}_1[(\rho^T \otimes I_m)C_\Phi]$.

Theorem 3.6 Under the Choi-Jamiołkowski isomorphism, Φ is completely positive if and only if $C_\Phi \succeq 0$ [5], and Φ is trace-preserving if and only if $\text{tr}_2[C_\Phi] = I_n$.

Block positivity is the operator counterpart of map positivity: a map Φ is positive if and only if its Choi matrix C_Φ is block positive, exactly as it is completely positive if and only if $C_\Phi \succeq 0$.

The partial transpose is itself the ampliation of the transposition map T : $\rho^\Gamma = (I_m \otimes T)(\rho)$. Writing this out for the 2×2 case we get its Choi matrix:

$$C_T = \sum_{i,j=1}^2 E_{ij} \otimes E_{ij}^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (5)$$

with eigenvalues ± 1 . Since $C_T \not\succeq 0$, the transpose is *not* completely positive even though it is positive; it is the prototypical PNCP map.

A word to the wise: the Choi matrix of the composition $\Phi \circ \Psi$ is not the product $C_\Phi C_\Psi$. Instead, it is obtained as the ampliation $I_n \otimes \Phi$ of C_Ψ :

$$C_{\Phi \circ \Psi} = (I_n \otimes \Phi)(C_\Psi). \quad (6)$$

Expanding in the standard basis yields the index formula

$$(C_{\Phi \circ \Psi})_{ip,jq} = \sum_{k,l} (C_\Psi)_{ik,jl} (C_\Phi)_{kp,lq}. \quad (7)$$

This formula is used directly in the implementation to compose two candidate PPT maps from their Choi matrices.

3.4 Entanglement

A *quantum state* on $\mathbb{C}^n$ is a density matrix $\rho \in M_n^+$ with $\text{tr}(\rho) = 1$. A *quantum channel* is a completely positive trace-preserving (CPTP) map $\Phi : M_n \rightarrow M_m$.

Definition 3.7 (*Separability and entanglement*) A bipartite state $\rho \in M_m \otimes M_n$ is *separable* if

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \rho_i^A \in M_m^+, \quad \rho_i^B \in M_n^+. \quad (8)$$

Otherwise ρ is *entangled*.

Definition 3.8 (*Entanglement-breaking map*) A map $\Phi : M_n \rightarrow M_m$ is *entanglement breaking* (EB) if $(I_k \otimes \Phi)(\rho)$ is separable for every $k \in \mathbb{N}$ and every state $\rho \in M_k \otimes M_n$ [1].

Testing $k = 1$ alone would only check that Φ preserves separability of bipartite states; the ampliation over all k ensures Φ destroys entanglement with any external reference system. Equivalently, Φ is EB if and only if its Choi matrix C_Φ is a separable state [1].

Definition 3.9 (*PPT state and PPT map*) A state $\rho \in M_m \otimes M_n$ is *PPT* if $\rho \succeq 0$ and $\rho^\Gamma \succeq 0$. A map $\Phi : M_n \rightarrow M_m$ is *PPT* if its Choi matrix is PPT:

$$C_\Phi \succeq 0 \quad \text{and} \quad C_\Phi^\Gamma \succeq 0. \quad (9)$$

Since $C_\Phi \succeq 0$ characterizes CP maps, a PPT map is a CP map whose Choi matrix has non-negative partial transpose. The composition of two PPT maps is trivially PPT.

3.4.1 PPT2 Conjecture

With PPT maps (Definition 3.9) and entanglement-breaking maps (Definition 3.8) in hand, we can now state the central object of study.

Theorem 3.10 (*PPT2 Conjecture*) If Φ_1 and Φ_2 are PPT maps, then $\Phi_1 \circ \Phi_2$ is entanglement breaking [2].

The conjecture is proven for $n = 2$ (since all PPT states in $M_2 \otimes M_n$ are separable [6]), for $n = 3$ [3], and for Choi-type maps in all dimensions [4]. It remains open for $n \geq 4$ in the general case. The rest of this chapter develops the formalism needed to study the conjecture computationally.

3.5 Testing entanglement

Deciding separability is NP-hard in general [7]. Since an exact test is computationally intractable, one relies on necessary conditions: a state that fails such a condition must be entangled. However, satisfying all known criteria does not guarantee separability. For a comprehensive survey see [8].

3.5.1 The PPT test

If a state ρ is separable, i.e. $\rho = \rho_A \otimes \rho_B$, then it is also PPT, since $\rho^\Gamma = \rho_A \otimes \rho_B^T \succeq 0$. Furthermore, the transposition map T is positive but not completely positive, applying $I \otimes T$ to an entangled state can produce a non-PSD result. This means that if $\rho^\Gamma \not\succeq 0$, then it must be entangled. The condition is necessary but not sufficient: bound entangled (PPT entangled) states exist for systems with $mn > 6$ [1].

3.5.2 Positive maps and entanglement witnesses

The set of separable states SEP is convex and closed. By the Hahn-Banach separation theorem; any point outside a closed convex set is separated from it by a hyperplane, for every entangled $\rho \notin \text{SEP}$ there exists a Hermitian operator W (a hyperplane in H_{mn}) with $\text{tr}[W\rho] < 0$ and $\text{tr}[W\sigma] \geq 0$ for all $\sigma \in \text{SEP}$ [1].

Definition 3.11 (*Entanglement witness*) A Hermitian operator $W \in H_{mn}$ is an *entanglement witness* if $\text{tr}[W\sigma] \geq 0$ for every separable state $\sigma \in M_m \otimes M_n$ and $\text{tr}[W\rho] < 0$ for some entangled state $\rho \in M_m \otimes M_n$.

Geometrically, each witness W cuts off a half-space that contains ρ but not SEP ; the intersection of all such half-spaces recovers SEP exactly.

Witnesses are one-sided: $\text{tr}[W\rho] < 0$ certifies entanglement, but $\text{tr}[W\rho] \geq 0$ does not certify separability. An operator is a (valid) entanglement witness precisely when it is block positive but not PSD.

Definition 3.12 (*Decomposable witness*) A witness W is *decomposable* if $W = P + Q^\Gamma$ for some $P, Q \succeq 0$; otherwise it is *non-decomposable* [9].

A known result is that decomposable witnesses cannot detect PPT entangled states. For any PPT σ , we have

$$\text{tr}[(P + Q^\Gamma)\sigma] = \text{tr}[P\sigma] + \text{tr}[Q\sigma^\Gamma] \geq 0. \quad (10)$$

This means only non-decomposable witnesses are useful for our purposes.

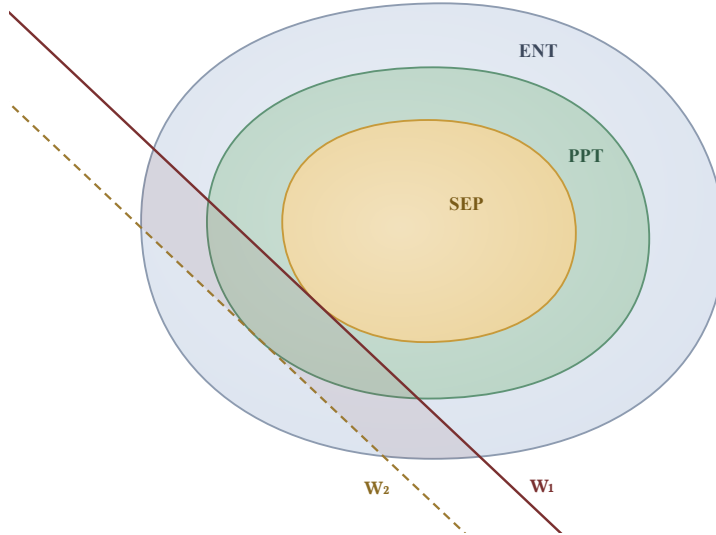


Figure 1: Geometric picture (schematic) of entanglement witnesses as separating hyperplanes over the nested convex sets of separable, PPT and entangled states. Inclusions are strict, with bound-entangled states populating the outside of the PPT region (green strip). The entanglement witness W_1 (solid) is *finer* than W_2 (dashed), it detects every state W_2 does, in addition to the states W_2 misses (shaded band).

Under the Choi-Jamiolkowski isomorphism, every entanglement witness $W = C_\Phi$ corresponds to a PNCP map Φ , and vice versa; block positivity of W is exactly positivity of Φ . This gives two ways to test a state. The scalar test asks whether $\text{tr}[C_\Phi \rho] < 0$, i.e. whether ρ lies on the negative side of the single separating hyperplane defined by C_Φ . The map test asks the more demanding question whether the ampliation $(I_k \otimes \Phi)(\rho)$ has *any* negative eigenvalue, $(I_k \otimes \Phi)(\rho) \not\geq 0$. The scalar test inspects only one expectation value and so can miss a negative eigenvalue in a direction other than the one C_Φ singles out; the map test inspects the whole spectrum. Concretely, $(I_k \otimes \Phi)(\rho) \not\geq 0$ holds exactly when some vector ψ satisfies

$$\langle \psi | (I \otimes \Phi)(\rho) | \psi \rangle = \text{tr}[(I \otimes \Phi^*)(|\psi\rangle\langle\psi|)\rho] < 0. \quad (11)$$

The map therefore encodes an entire family of scalar witnesses $W_\psi = (I \otimes \Phi^*)(|\psi\rangle\langle\psi|)$, one per output vector $|\psi\rangle$; the Choi-matrix test $\text{tr}[C_\Phi \rho]$ is the single member obtained from the maximally entangled $|\psi\rangle$, whereas the eigenvalue check optimizes over all $|\psi\rangle$ at once. Hence

$$\lambda_{\min}((I \otimes \Phi^*)(\rho)) \leq \text{tr}[C_\Phi \rho], \quad (12)$$

so the map detects every state the scalar test does, and in general strictly more [1].

The PPT test is the special case of this positive-map test in which $\Phi = T$ is the transposition map: $\rho^\Gamma \not\geq 0$ is exactly $(I \otimes T)(\rho) \not\geq 0$. The transpose is, however, a *decomposable* witness, so it sees only NPT entanglement; detecting PPT entanglement requires stronger maps.

3.5.3 Further separability criteria

Several other necessary conditions for separability are known; each can detect entanglement the PPT test misses, but none is sufficient.

Reduction criterion. If $\rho \in M_m \otimes M_n$ is separable, then

$$\rho_A \otimes I_n - \rho \succeq 0 \quad \text{and} \quad I_m \otimes \rho_B - \rho \succeq 0, \quad (13)$$

where $\rho_A = \text{tr}_B[\rho]$ and $\rho_B = \text{tr}_A[\rho]$ are the *reduced states* (obtained by tracing out one subsystem) [10].

Range criterion. If $\rho \in M_m \otimes M_n$ is separable, there exist product vectors $\{|\psi_i \otimes \varphi_i\rangle\}$ spanning the range of ρ such that $\{|\psi_i \otimes \overline{\varphi_i}\rangle\}$ spans the range of ρ^Γ [11]. The range criterion can detect certain PPT entangled states, but fails when ρ is full rank (e.g., under noise), since then any set of vectors spans its range trivially.

Majorization criterion. If ρ is separable, then $\lambda(\rho) \preceq \lambda(\rho_A)$ and $\lambda(\rho) \preceq \lambda(\rho_B)$, where $\lambda(\cdot)$ denotes the non-increasingly ordered eigenvalue vector

[12]. Note, here $a \preceq b$ has nothing to do with positive definiteness; it denotes *majorization*: $\sum_{i=1}^k a_i \leq \sum_{i=1}^k b_i$ for all k , with equality at $k = n$. This criterion follows from the reduction criterion [12] and therefore shares its limitations.

CCNR / realignment criterion. Define the *realignment* of ρ as the matrix $R(\rho)$ with entries $(R(\rho))_{i(m+k),j(n+l)} = \rho_{i(n+j),k(n+l)}$. If ρ is separable, then $\|R(\rho)\|_1 \leq 1$, where $\|\cdot\|_1$ is the trace norm [13]. The CCNR criterion is independent of the PPT criterion and can detect some PPT entangled states that the other criteria miss.

Beyond criterion-based tests, there are *algorithmic* approaches that reformulate separability as a convex optimization problem, i.e. the DPS hierarchy (section 3 The DPS hierarchy). There are also criteria based on covariance matrices and Bell inequalities that are less directly applicable in our setting, for a full survey see [8].

3.6 Semidefinite Programming

A *semidefinite program* (SDP) is a convex optimization problem in which a linear objective is minimized subject to a linear matrix inequality. The standard primal form is

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && F(x) := F_0 + \sum_{i=1}^n x_i F_i \succeq 0, \end{aligned} \tag{14}$$

where $x \in \mathbb{R}^n$ is the optimization variable, $c \in \mathbb{R}^n$ is the cost vector, and $F_0, F_1, \dots, F_n \in H_d$ are fixed Hermitian matrices. The associated *dual SDP* is

$$\begin{aligned} & \text{maximize} && -\text{tr}[F_0 Z] \\ & \text{subject to} && Z \succeq 0, \\ & && \text{tr}[F_i Z] = c_i, \quad i = 1, \dots, n. \end{aligned} \tag{15}$$

The *Slater condition* for (14) requires the existence of a strictly feasible point: some x with $F(x) \succ 0$. When it holds for both primal and dual, *strong duality* holds: primal and dual optima coincide. When $c = 0$, (14) is a *feasibility problem*. If infeasible, a dual feasible $Z \succeq 0$ with $\text{tr}[F_i Z] = 0$ and $\text{tr}[F_0 Z] > 0$ certifies infeasibility.

3.6.1 Interior point methods

SDPs are solved in practice by *interior point methods* (IPMs), which follow a smooth trajectory through the strict interior of the feasible region. A standard IPM applied to a $d \times d$ SDP with n variables reaches ε -accuracy in $O(\sqrt{d} \log(1/\varepsilon))$ iterations, each requiring $O(n^2 d^2 + n d^3)$ operations [14].

This thesis uses MOSEK [15], a state-of-the-art IPM solver for semidefinite programs.

A property of IPMs important for our use: solutions lie in the *strict interior* of the feasible region $F(\mathbf{x}) \succ 0$. Since all solutions are floating-point approximations, this provides the numerical slack around the boundary that enables rationalization as a post solver method of certifying solutions.

3.6.2 Sum-of-squares relaxation

Semidefinite programming also underlies a general relaxation of polynomial non-negativity. Let $\mathbb{R}[\mathbf{x}, \mathbf{y}]$ be the ring of real polynomials in $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^m$. A polynomial p is *non-negative* if $p(\mathbf{x}, \mathbf{y}) \geq 0$ for all real inputs; it is a *sum of squares* (SOS) if $p = \sum_i q_i^2$ for polynomials q_i . Every SOS polynomial is non-negative, but not every non-negative polynomial is SOS.

Definition 3.13 (*Gram matrix representation*) Let $\mathbf{z} = (z_1, \dots, z_N) \in \mathbb{R}^N$ and let $\mathbf{v}(\mathbf{z})$ represent the $\binom{N+d-1}{d}$ monomials of degree d in $\mathbf{z}$. A homogeneous polynomial p of degree $2d$ is SOS if and only if there exists a symmetric PSD matrix $G \succeq 0$ (a *Gram matrix* of p), such that

$$p(\mathbf{z}) = \mathbf{v}(\mathbf{z})^T G \mathbf{v}(\mathbf{z}). \quad (16)$$

Testing whether p is SOS therefore reduces to a semidefinite feasibility problem: find $G \succeq 0$ satisfying the linear constraints that equate coefficients of (16) with those of p . If no such G exists, p is not SOS.

Testing the nonnegativity of a polynomial p is an NP-hard problem [16]. The standard way around this is to use SOS relaxations which are computationally tractable. The idea is to multiply p by a fixed SOS multiplier and test the product for SOS.

3.6.3 Entanglement testing as an SDP

We can rephrase entanglement testing as maximizing $\text{tr}[M\rho]$, where $\rho \in \text{SEP}$. This is a convex optimization problem, but the set SEP is not semidefinite representable. Therefore, we find tractable outer approximations of SEP , to get (a sequence of) necessary conditions for separability. Each approximation is defined by semidefinite constraints, so membership can be tested by an SDP feasibility problem. If ρ fails the test at some level, we can conclude it must be entangled.

reference, explanation of Positivstellensatz, formula for the multiplier/retrieving polynomial from the multiplier

3.6.4 The DPS hierarchy

The Doherty-Parrilo-Spedalieri (DPS) hierarchy [17] provides such a sequence of SDP relaxations of separability, with the added benefit of being *complete*: for every entangled state there exists a finite level k at which

the test detects it. In principle, running the hierarchy to convergence solves separability exactly; in practice, only the first few levels are computationally feasible.

Symmetric extensions. For a separable state $\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$, define the k -fold *symmetric extension*

$$\rho_k = \sum_i p_i \rho_i^A \otimes (\rho_i^B)^{\otimes k} \in H(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k}). \quad (17)$$

This extension satisfies three properties, each with a physical interpretation:

$$(I_m \otimes \Pi_k) \rho_k (I_m \otimes \Pi_k) = \rho_k, \quad (18)$$

$$\rho_k^{\Gamma_s} \succeq 0 \quad \forall s = 1, \dots, k, \quad (19)$$

$$\text{tr}_{B_2 \dots B_k} [\rho_k] = \rho. \quad (20)$$

Here Π_k projects onto the bosonic (fully symmetric) subspace of $(\mathbb{C}^n)^{\otimes k}$. Condition (18) states that ρ_k is invariant under permutations of the k copies of B . Condition (19) states that all partial transposes of ρ_k are PSD. Condition (20) states that tracing out the extra $k - 1$ copies of B recovers ρ .

Definition 3.14 (*DPS set at level k*) DPS_n^k is the set of states $\rho \in H(\mathbb{C}^m \otimes \mathbb{C}^n)$ for which there exists a symmetric extension $\rho_k \in H(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k})$ satisfying (18), (19), and (20).

Each DPS_n^k is defined by semidefinite constraints, so membership is testable in polynomial time for fixed k . The hierarchy satisfies [17]:

1. $\text{SEP}_n \subseteq \text{DPS}_n^k$ and $\text{DPS}_n^{k+1} \subseteq \text{DPS}_n^k$ for all $k \geq 1$.
2. DPS_n^1 is equivalent to the PPT criterion.
3. Asymptotic completeness: $\bigcap_{k \geq 1} \text{DPS}_n^k = \text{SEP}_n$.

Feasibility and witness extraction. Testing $\rho \in \text{DPS}_n^k$ amounts to searching for ρ_k satisfying (18), (19), and (20): this is an SDP feasibility problem. When it is infeasible (no valid extension exists), the dual variable $Z \succeq 0$ yields an entanglement witness. Concretely, the SDP dual to the level- k test has a feasible Z whenever ρ is entangled at that level. The operator

$$W = \text{tr}_{B_2 \dots B_k} [Z] \quad (21)$$

is an entanglement witness for ρ [17], i.e. $\text{tr}[W\rho] < 0$ verifies entanglement directly.

Improvements and limitations. Several enhancements to the basic hierarchy are known. Harrow, Natarajan, and Wu [18] add first-order optimality (KKT) conditions to the DPS SDP, achieving *finite convergence*: infeasibility is certified at a finite level rather than only asymptotically. The KKT

conditions are linear constraints on the Lagrange multipliers of the original SDP; they increase the variable count substantially and make clean witness extraction from the dual more involved. Specialized hierarchies have been developed for states with symmetry. For diagonal unitary invariant states, Britz and Laurent [19] give a drastically smaller SDP at each level. For Werner states and isotropic states, explicit separability conditions are known. However, none of these apply to our search: the PPT2 conjecture is already proven for the relevant symmetric families [4], so our search must use the general DPS hierarchy.

3.6.5 Maps as polynomials

An alternative approach to detecting entanglement exploits the polynomial representation of linear maps. Each linear map $\Phi : M_n \rightarrow M_m$ corresponds to a biquadratic polynomial:

$$p_\Phi(\mathbf{x}, \mathbf{y}) := \mathbf{y}^T \Phi(\mathbf{x}\mathbf{x}^T) \mathbf{y}. \quad (22)$$

The fundamental correspondence [20] between the map and polynomial representation is:

- Φ is *positive* if and only if p_Φ is non-negative on $\mathbb{R}^n \times \mathbb{R}^m$.
- Φ is *completely positive* if and only if p_Φ is SOS.

A PNCP map corresponds exactly to a non-negative non-SOS polynomial. Each such a polynomial therefore gives rise to an entanglement witness, and the SOS test of the previous section becomes a test of complete positivity. This is the route we use to manufacture entanglement witnesses directly.

3.6.5.1 KMSZ construction for PNCP maps

Now that we have established the value of non-negative non-SOS polynomials for our purposes, we present a construction [20] that produces such polynomials, and therefore PNCP maps, as follows:

1. Sample random points $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(t)} \in \mathbb{R}^n$ and $\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(t)} \in \mathbb{R}^m$.
2. Form bilinear $h_j(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}^{(j)}, \mathbf{x} \rangle \cdot \langle \mathbf{y}^{(j)}, \mathbf{y} \rangle$, each a product of two linear forms, so $\sum_j h_j^2$ is SOS.
3. Find $f \notin \text{span}\{h_1, \dots, h_t\}$ such that f is not SOS.
4. Find $\delta > 0$ small enough that $F_\delta := \delta f + \sum_j h_j^2 \geq 0$.

Clearly F_δ is by construction *not* SOS, so all that is left is to verify the non-negativity. Steps 1–3 use only linear algebra, and step 4 involves solving an SDP. However, we cannot directly represent F_δ with semidefinite constraints. Instead, we relax the condition and search for a Gram matrix $G \succeq 0$ for $F_\delta \cdot S$, where $S = \sum_{i,j} x_i^2 y_j^2$ is a fixed SOS multiplier. This bihomogeneous multiplier preserves the separate degrees in $\mathbf{x}$ and $\mathbf{y}$, matching the biqua-

dratic structure of F_δ . If feasible, then $F_\delta \cdot S$ is SOS, which certifies that F_δ is non-negative [20].

3.6.5.2 Real maps and their complexification

The correspondence above is stated for real symmetric inputs: $p_\Phi(\mathbf{x}, \mathbf{y}) = \mathbf{y}^T \Phi(\mathbf{x}\mathbf{x}^T) \mathbf{y}$ probes Φ only on rank-one *real* symmetric matrices $\mathbf{x}\mathbf{x}^T$, and “non-negative” means $p_\Phi \geq 0$ on $\mathbb{R}^n \times \mathbb{R}^m$. Accordingly, the KMSZ construction produces a *real* linear map

$$\Phi : S_n(\mathbb{R}) \rightarrow S_m(\mathbb{R}). \quad (23)$$

Quantum states, however, are complex Hermitian operators, so to use Φ as an entanglement witness it must be extended to a complex map. Following [20], we take the *complexified trivial extension*

$$\Gamma_\mathbb{C} := (\Phi \oplus 0)_\mathbb{C} : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C}), \quad (24)$$

defined in two steps. First, using $M_n(\mathbb{R}) = S_n(\mathbb{R}) \oplus K_n(\mathbb{R})$, extend Φ to all real matrices by acting as zero on the skew-symmetric part:

$$(\Phi \oplus 0)(S + K) = \Phi(S), \quad S \in S_n(\mathbb{R}), \quad K \in K_n(\mathbb{R}). \quad (25)$$

Then extend $\dagger$ -linearly to complex matrices,

$$\Gamma_\mathbb{C}(A + iB) = (\Phi \oplus 0)(A) + i(\Phi \oplus 0)(B), \quad A, B \in M_n(\mathbb{R}), \quad (26)$$

the unique extension satisfying $\Gamma_\mathbb{C}(X^*) = \Gamma_\mathbb{C}(X)^*$. Klep et al. prove that $\Gamma_\mathbb{C}$ is positive (respectively PNCP) whenever Φ is [20].

Two observations make this extension harmless to compute with. First, for a Hermitian input $\rho = A + iB$ (with $A \in S_n(\mathbb{R})$ symmetric and $B \in K_n(\mathbb{R})$ skew-symmetric, since $\rho = \rho^*$) only the symmetric part survives, $\Gamma_\mathbb{C}(\rho) = \Phi(A)$. Second, and equivalently, the Choi matrix of $\Gamma_\mathbb{C}$ is exactly the real symmetric matrix C_Φ returned by the construction, now read as a complex Hermitian operator; no entries change. A by-product of annihilating the skew part is that C_Φ is *partial-transpose invariant*, $C_\Phi^\Gamma = C_\Phi$, a property we return to in section 5 Representation and symmetry effects.

3.6.5.3 Indecomposability of the generated maps

For complex maps the appropriate polynomial object is *bi-Hermitian*: with $z \in \mathbb{C}^n$, $w \in \mathbb{C}^m$ and their conjugates, the polynomial

$$p_{\Gamma_\mathbb{C}}(z, w) = \langle w | \Gamma_\mathbb{C}(|z\rangle\langle z|) | w \rangle \quad (27)$$

is real-valued, and it is non-negative on all product vectors if and only if $C_{\Gamma_\mathbb{C}}$ is block-positive, i.e. an entanglement witness. For such polynomials there are *two* inequivalent notions of sum-of-squares [21]:

- *complex SOS* (CSOS): $p = \sum_i |q_i(z, w)|^2$ with each q_i holomorphic (a polynomial in z, w , not $\bar{z}, \bar{w}$);
- *real SOS* (RSOS): $p = \sum_i g_i^2$ with each g_i a Hermitian (real-valued) polynomial in $z, \bar{z}, w, \bar{w}$.

CSOS is a strictly stronger condition than RSOS. Under the map correspondence [21], a CSOS certificate corresponds exactly to a completely positive map and an RSOS certificate to a *decomposable* witness. Failure of any RSOS decomposition is therefore the polynomial signature of indecomposability, and by Definition 3.12 such a witness can detect PPT entangled states.

The two representations are linked by a structural identity. Writing $z = a + ib$ and $w = c + id$ with $a, b \in \mathbb{R}^n$ and $c, d \in \mathbb{R}^m$, a direct expansion gives

$$p_{\Gamma_{\mathbb{C}}}(a + ib, c + id) = p_{\Phi}(a, c) + p_{\Phi}(b, c) + p_{\Phi}(a, d) + p_{\Phi}(b, d). \quad (28)$$

The complexified polynomial is thus completely controlled by the values of the real form p_{Φ} on real inputs, which can be used to prove *indecomposability* [22].

Theorem 3.15 (*Indecomposability*) If p_{Φ} is not SOS, then $p_{\Gamma_{\mathbb{C}}}$ is not RSOS, and hence the witness $C_{\Gamma_{\mathbb{C}}}$ is non-decomposable.

Since p_{Φ} is bihomogenous it vanishes whenever either argument is zero, so restricting to the real slice, i.e. setting $b, d = 0$ we find (28) collapses to $p_{\Gamma_{\mathbb{C}}}(a, c) = p_{\Phi}(a, c)$ and $p_{\Gamma_{\mathbb{C}}} \in \text{RSOS}$ would imply $p_{\Phi} \in \text{SOS}$.

Because the KMSZ form p_{Φ} is non-SOS by construction, every generated witness is indecomposable, and therefore capable of detecting some PPT entangled state. This settles, at least in principle, our concerns about the efficacy of using such maps for our purposes.

Knowing the generated witnesses *can* detect PPT entanglement is not our only concern; we also need to know evaluate their detection power. This is determined by where they lie in the space of entanglement witnesses.

We say a witness W is *tight* [23] if there exists some product vector $x \otimes y$ for which $\langle x \otimes y | W | x \otimes y \rangle = 0$, geometrically this means the supporting hyperplane genuinely touches SEP. Further, W is *optimal* [9] if there is no $P \succeq 0$ such that $W - P$ remains block-positive; i.e. *tilting* the hyperplane toward the SEP. By construction, the generated witnesses are tight: the algorithm prescribes zeros at the sampled Segre points, so the hyperplane defined by W touches SEP at those points. Optimality however, is a stronger condition, and as of yet undetermined for these witnesses [22]. A practical consequence is that the witnesses lie close to the decomposable cone, and

while they are robust in detecting NPT entanglement, they detect PPT entanglement only weakly.

3.6.5.4 Non-uniqueness of the matrix representation

The polynomial p_Φ does not determine the matrix C_Φ uniquely. Writing p_Φ as a quadratic form in the product monomials $\mathbf{z} = \mathbf{x} \otimes \mathbf{y}$,

$$p_\Phi(\mathbf{x}, \mathbf{y}) = \mathbf{z}^T M \mathbf{z}, \quad (29)$$

we see any symmetric M that reproduces the coefficients of p_Φ is an admissible representation. Two such matrices M, M' give the same polynomial precisely when $\mathbf{z}^T (M - M') \mathbf{z} = 0$. The matrices with this property form a linear space L , spanned by the 2×2 minor (*Segre*) relations

$$(x_i y_k)(x_j y_l) - (x_i y_l)(x_j y_k) = 0, \quad i < j, \quad k < l. \quad (30)$$

There is one independent relation for each choice of two rows $i < j$ and two columns $k < l$, so $\dim L = \binom{n}{2} \binom{m}{2}$. The admissible representations therefore form an affine space $M_0 + L$.

The important distinction is that the relations defining L vanish on *real* product vectors, not on complex ones, so members of $M_0 + L$ are not necessarily valid witnesses or even block-positive over $\mathbb{C}$. The canonical choice is the symmetric, partial-transpose-invariant representative (24), where zeroing the skew-symmetric part guarantees positivity. Whether the skew-symmetric directions can be used while preserving positivity over $\mathbb{C}$ remains open.

3.6.5.5 Rationalization

It is possible to sample over $\mathbb{Q}$ to produce a rational polynomial, but the final step in our construction relies on solving an SDP, which inherently works with floating-point arithmetic. For an exact certificate it is therefore more prudent to rationalize only the final result.

Let G be a numerical solution with $\mu = \min(\text{eig}(G_0)) > 0$ and residual $\varepsilon = \max_i |\langle A_i, G_0 \rangle - b_i|$, then for $\mu > \varepsilon$, a rational feasible $\hat{G}$ can be obtained [24], [25] in two steps.

1. Compute a rational approximation $\hat{G}$ of the gram matrix G satisfying $\|\hat{G} - G_0\|^2 + \varepsilon^2 < \mu^2$.
2. Project $\hat{G}$ back to the affine subspace given by the constraints.

With a rational gram matrix $\hat{G}$ in hand, the coefficients of $\hat{F}_\Phi$ can be isolated using exact computations.

4 Methods

This chapter defines the computational workflow used in the current implementation. The goal is reproducibility with mathematical traceability. Each computational step corresponds to a defined object or operation from the preceding chapter.

A note on normalization is in order. A quantum state is conventionally a PSD operator with unit trace, and a quantum channel is trace preserving. The tests in this thesis are, however, positivity and sign based. Detecting entanglement amounts to checking the sign of a partial transpose, a witness expectation, or an SOS margin, none of which depend on the magnitude of the trace. We therefore work throughout with unnormalized operators, and normalize only when necessary. This avoids redundant rescaling in the inner loops without affecting any conclusion.

Two design constraints shape the workflow:

1. *exact* separability testing is intractable in the dimensions where the conjecture is open (3 The DPS hierarchy), partial testing is very much doable, but every complete test is a one-sided relaxation;
2. SDP solutions are floating-point and fragile near feasibility boundaries, so positive detections are re-validated exactly (3 Rationalization).

4.1 Generating a witness library

Stage 1 builds the witness library. We run the KMSZ construction to produce N_W PNCP maps, and rationalize each certificate so that every stored witness is exact, guarding against the false positives that floating-point SOS tests invite near the boundary.

The rationalization, developed in section 3 Rationalization, proceeds as follows:

1. Extract the Gram matrix G from the SDP constraints,

$$(\delta f + h^2) \left(\sum_{i,j} x_i^2 y_j^2 \right)^l = (\mathbf{x}\mathbf{y})^T G \mathbf{x}\mathbf{y} \quad (31)$$

2. Eigendecompose G exploiting the known e -dimensional nullspace [20]; set the first $e = (n-1)(m-1)$ eigenvalues to zero, and rationalize the remaining coefficients to obtain $\hat{G}$.

3. Reformulating the constraints in terms of $\hat{G}$, isolate the coefficients of the polynomial $\hat{p} = \delta f + h^2$ from the relaxation terms $\left(\sum_{i,j} x_i^2 y_j^2\right)^l$.
4. Additionally we solve a final feasibility SDP to double check we have not made $\hat{p}$ into a SOS.

Upon completion we are confident in the validity of the certificate, so we can choose to represent it in floating point or as a rational number, as long as we are sure to use high enough precision. Typically, solvers work with a tolerance of $\approx 10^{-8}$.

4.2 Generating candidates

We generate PPT states directly as the Choi matrices of PPT maps. A short sampling routine produces a random PPT state; an entanglement filter then keeps only the candidates that can plausibly yield a counterexample. A second, witness-guided generator targets bound entangled states directly.

4.2.1 Random PPT states

Algorithm 1: Generating a random PPT state

- 1: Sample a real matrix $R \in M_{nm}$ with i.i.d. standard-normal entries and form the PSD matrix $\rho = RR^T$.
 - 2: Compute the smallest eigenvalue of the partial transpose, $\lambda = \lambda_{\min}(\rho^\Gamma)$. If $\lambda < 0$, set $\rho \leftarrow \rho - \lambda I$.
-

The construction is correct without any further work. By definition $\rho = RR^T \succeq 0$, and when $\lambda < 0$ the shift adds the non-negative multiple $|\lambda| I$ to ρ , so positivity is preserved. The same shift acts on the partial transpose as $\rho^\Gamma - \lambda I$, because $I^\Gamma = I$, which raises every eigenvalue of ρ^Γ by $|\lambda|$ and hence makes $\rho^\Gamma \succeq 0$. The resulting ρ is therefore PPT. Sampling element-wise from the normal distribution produces a representative collection of states [26].

The off-diagonal blocks may optionally be symmetrized before the shift, for each block pair (i, j) with $i < j$, both ρ_{ij} and ρ_{ji} are replaced by their average $(\rho_{ij} + \rho_{ji})/2$, which forces $\rho = \rho^\Gamma$, invariant under partial transpose (IPT). This symmetrization is not essential. We only check if it makes the computation any simpler or faster, since a single positivity shift then certifies both $\rho \succeq 0$ and $\rho^\Gamma \succeq 0$ at once. The Choi matrix of a map is only defined up to the basis convention, so restricting to such IPT representatives is a permissible convenience rather than a requirement. The motivation is that our precomputed witnesses are themselves generated in this IPT shape, so matching the candidates to the same form may make them easier to detect; we therefore run the search in two versions, with and without symmetry.

A random PPT state may be separable or bound entangled, and only the entangled ones are useful here. If either composed map is entanglement breaking, equivalently, has a separable Choi matrix, then the composition is automatically entanglement breaking and cannot violate the conjecture. We therefore discard separable candidates and keep only states for which we can certify entanglement, assembling a pool of genuine bound entangled states whose compositions are worth testing.

4.2.2 Bound entangled states from a witness

The random sampler has no control over entanglement, so many of its draws are discarded. A second generator instead targets entanglement directly, using the witness library to manufacture states it is guaranteed to detect. Fix a witness W and minimize its expectation over the whole PPT cone,

$$\begin{aligned} & \text{minimize} && \text{tr}[W\rho] \\ & \text{subject to} && \rho \succeq 0, \\ & && \rho^\Gamma \succeq 0, \\ & && \text{tr}[\rho] = 1. \end{aligned} \tag{32}$$

This is a single SDP. Since every separable state gives $\text{tr}[W\rho] \geq 0$, a negative optimum exhibits a PPT *entangled* state that W detects, one bound entangled candidate per witness that admits one. The optimum also measures the witness's detection strength, and is a ready-made hard instance for the following composition tests.

4.2.3 Alternative bound-entangled constructions

The first method, i.e. random sampling is fast, but it gives no control over entanglement. So we must filter states out, which necessitates solving an SDP. Not a great option for scalable searches. The second method is more targeted, but still suffers from the same problem, though the SDP is much smaller in this case. There are several constructions in the literature that produce bound entangled states by design, however we opted not to use them in the search, for various reasons noted below. They are however, the natural starting points for further investigation.

- **Unextendible product bases (UPB)** are a set of mutually orthogonal product vectors spanning a proper subspace whose orthogonal complement contains no product vector. The normalized projector onto that complement is PPT and entangled [27]. UPBs give explicit, low-rank bound entangled states, but the construction is the opposite of generic: it relies on specific, hand-picked bases that cannot be sampled at random,

and in some dimensions may not exist at all, so it cannot drive a randomized search.

- **Antisymmetric-subspace states** are a simple class of PPT entangled states from the projectors onto the symmetric and antisymmetric subspaces of two identical systems, generalizing the Werner states [28]. The construction is explicit, but certifying entanglement of the resulting states still reduces to an SDP, so it does not scale better than our pipeline.
- **Symmetric random induced states** are a more recently studied class of bound entangled states [29]. Sampling a random pure state from the symmetric subspace $\mathcal{H}_S^{N+N_A+1}$ and tracing out N_A subsystems, leaving a mixed state on $\mathcal{H}_S^{N+1}$ produces, with high probability, a bound entangled state. The catch is dimensionality: for $N = 4$ bound entanglement is most likely at $N_A = 12$, which demands an enormous ancillary space and correspondingly heavy computation. Even then the probability of entanglement stays below 0.5, comparable to the hit rate of our own construction. It is nonetheless a promising state-of-the-art source of candidates and a useful point of comparison.

4.3 Testing candidates

Given two PPT candidates Φ_1, Φ_2 from the pool, we form the Choi matrix of their composition using the ampliation operation (6). Then we test the composite for entanglement: if any composition is ever found to be entangled, the PPT2 conjecture is violated. Because composition is not commutative, the search ranges over *every ordered pair* of pool states, self-pairs included.

4.3.1 Composition search with the witness library and DPS

Each composite is screened by three complementary detection tests:

- **Scalar witness test.** For each witness W in the precomputed library, $\text{tr}[WC] < 0$ certifies that C is entangled.
- **Map witness test.** The *stronger* condition $(I \otimes \Phi_W)(C) \not\geq 0$, evaluated as a smallest-eigenvalue check on the ampliation of the witness map; this can fire where the scalar test does not.
- **DPS relaxation.** The level-2 DPS relaxation; a feasible dual certificate flags entanglement.

The witness library and the DPS test are complementary: precomputed witnesses are cheap to evaluate but are individually much less likely to detect entanglement, whereas the DPS relaxation is more robust but far more expensive. Running all three and recording each score also lets us compare the methods directly, how often the witnesses alone suffice versus how much the DPS relaxation adds (section 5 Results).

The components above assemble into a three-stage pipeline. The first two stages build two pools independently, a library of PNCP witness maps and a pool of bound entangled PPT states, and the third stage tests all of their compositions. Separating generation from testing lets each pool be built once, checkpointed, and reused across runs.

Algorithm 2: PPT2 counterexample search

```

1: ▷ Stage 1: witness library
2: for  $i = 1, \dots, N_W$  do
3:   Construct a PNCP map via 3 KMSZ construction for PNCP maps,
   rationalize its certificate (section 3 Rationalization), and store the
   witness  $W_i$ .
4: end
5: ▷ Stage 2: bound entangled candidate pool
6: for  $j = 1, \dots, N_C$  do
7:   Sample a random PPT state via Algorithm 1; keep it only if we can
   verify it is entangled.
8: end
9: ▷ Stage 3: test all compositions
10: for every ordered pair  $(a, b)$  do
11:   Form the composite channel  $C = (I_n \otimes \Phi_a)(C_{\Phi_b})$ .
12:   Flag  $C$  if any witness fires ( $\text{tr}[W_i C] < 0$  or  $(I \otimes \Phi_{W_i})(C) \neq 0$ ) or
   the level-2 DPS relaxation reports entanglement.
13:   If  $C$  is entangled, re-validate exactly and export the certificate: a
   counterexample to the PPT2 conjecture.
14: end

```

4.3.2 Direct PPT2 search by see-saw

The pipeline above tests a finite pool of compositions. We can instead search the composition manifold directly, asking whether a witness W can be made to fire anywhere on it. Sharpening (32), we minimise W 's expectation not over the whole PPT cone but over states that are themselves compositions of two PPT maps:

$$\begin{aligned}
& \text{minimize} && \text{tr}[W \cdot C_{\rho_1 \circ \rho_2}] \\
& \text{subject to} && \rho_1, \rho_2 \text{ PPT, unit trace.}
\end{aligned} \tag{33}$$

A composition of PPT maps is itself PPT, so a negative optimum would exhibit a PPT *and* entangled composition, a PPT2 counterexample witnessed by W . The ampliation is bilinear in the pair (ρ_1, ρ_2) , so (33) is not a single SDP but a bilinear matrix inequality; we solve it by *see-saw*, freezing one factor and optimising the other in alternation, restarting from several random PPT initialisations. Being non-convex, see-saw reaches only a local optimum,

so a non-negative result is supporting evidence rather than proof; the PPT-cone problem (32) is its convex relaxation and lower bound.

4.3.3 A note on the representation freedom

As a by-product we explored the representation freedom of section 3 Non-uniqueness of the matrix representation. Each witness can be expanded into a family of Gram representatives $M_0 + \sum_{\alpha} \lambda_{\alpha} N_{\alpha}$ over a basis $\{N_{\alpha}\}$ of L . We do not, in the end, rely on this. The relations spanning L vanish on *real* product vectors only, so an asymmetric representative stays block positive over $\mathbb{R}$ but need not remain so over $\mathbb{C}$. Guaranteeing such behavior is to our knowledge also an NP-hard problem without a simple solution in sight. The canonical, IPT representative, annihilating the skew-symmetric part as previously proposed remains the preferred approach for our search, together with generic, non-symmetrized candidates. We leave a principled complex-domain extension to future work.

4.4 Implementation Architecture

The implementation is a small Julia package in the `code/` directory. The core logic lives in the `ppt2` module (`code/src/ppt2.jl`), with the PNCP construction split into `code/src/pncp.jl` and included into the same module. Together they map one-to-one onto the operations defined above:

Function	Role
<code>rand_ppt</code>	Sample a random PPT state (Algorithm 1): the positivity shift, with optional block symmetrization (<code>ppt_invariant</code>).
<code>ampliation</code>	Compute $(I \otimes \Phi)(C)$, used both to compose maps and to apply a witness test.
<code>sample_pncp_form</code> , <code>segre_kernel_basis</code> , <code>non_sos_form</code>	The KMSZ construction: sample Segre-variety points, build the linear forms h_j , and produce the non-SOS quadratic form f .
<code>solve_sos</code>	Set up and solve the SOS feasibility/optimization SDP for a given relaxation degree l ; optionally trigger rationalization.
<code>rationalize_certificate</code>	Post-solver rationalization: zero the first e Gram eigenvalues, recover rational coefficients, and re-check non-SOS.
<code>find_pncp_poly</code> , <code>pncp_mat</code> , <code>poly2mat</code>	Orchestrate witness generation with retries and export the certificate as a Choi matrix.

Function	Role
<code>min_ppt_witness</code> , <code>min_ppt2_witness</code>	The witness-restricted SDPs (32) and (33): bound entangled states from a witness, and the see-saw search over compositions.

The module also exports a few supporting primitives: `rand_sep` and `rand_psd` for reference states, `is_ppt` for the PPT check, `gram_freedom` and `is_block_positive` for the witness-representation freedom and the block-positivity check of section 3 Non-uniqueness of the matrix representation, and `swap/antisymmetric_projector` for the antisymmetric-subspace construction of [28].

The package leans on the established Julia optimization and quantum-information stack rather than reimplementing it. Polynomials and the SOS cone are handled by `DynamicPolynomials` and `SumOfSquares`; the resulting semidefinite programs are modelled with `JuMP` and solved by `MOSEK` through `MosekTools` (any `JuMP`-compatible SDP solver could be substituted). The DPS hierarchy is not reimplemented: the search driver calls `entanglement_robustness` from `Ket`, an existing quantum-information toolbox that also supplies utilities such as the partial transpose. Matrices and witness libraries are serialized with `JLD2` so that generation and search can be separated and resumed.

Several command-line drivers in `code/scripts/` orchestrate the long-running jobs, all sharing a `common.jl` harness that provides resumable, reproducible, multithreaded batch generation: completed batches are detected and skipped on a rerun, and every candidate is seeded deterministically so a configuration yields the same dataset regardless of thread count. The generation drivers build the two pools: `gen_pncp.jl` constructs the PNCP witness library (Stage 1), while `gen_ppt.jl` samples and DPS-filters a random bound entangled candidate pool and `gen_witness_ppt.jl` produces, for each witness, the bound entangled state the witness-restricted SDP extracts from the full PPT cone (Stage 2). `test_ppt2.jl` then runs the threaded all-pairs search of Algorithm 2 (Stage 3), logging any detection together with the offending state and witness. The remaining drivers support the analysis of section 5 Results: `compare_detection.jl` records every criterion’s score on each detected state, the two witness tests and the DPS robustness, so the detection methods can be compared directly; `gen_witness_ppt2.jl` sharpens the witness-restricted SDP from the whole PPT cone down to the composition manifold by a see-saw over PPT-map pairs; and `cross_trace.jl` and `cross_ampl.jl` measure how broadly each witness detects states beyond its own. The `code/test/`

suite checks the construction against reference values and verifies that generated maps are positive on large random samples, and the `code/notebooks/` directory documents the rationalization, PPT-state, and UPB workflows interactively.

5 Results

This chapter summarizes computational results from the pipeline implementations.

5.1 Search scale and the broad composition search

All experiments are in the 4×4 case, the smallest dimension where the PPT2 conjecture remains open; the methods extend to slightly larger dimensions, but the cost grows quickly and we did not pursue them.

Stage 1 produced a library of $\langle 10,000 \rangle$ PNCP witness maps via the KMSZ construction (3 KMSZ construction for PNCP maps), each verified positive but not completely positive through the rationalization and SOS checks of section 3 Rationalization. Stage 2 produced the bound entangled candidate pool in two ways: random sampling with a level-2 DPS entanglement filter (Algorithm 1) kept $\langle 849 \rangle$ of $\langle 1,435 \rangle$ sampled states (a hit rate of $\langle 59\% \rangle$), and the witness-restricted SDP (32) extracted a further $\langle 10,000 \rangle$ bound entangled states, one per detecting witness. Building the two pools took approximately $\langle 5 \rangle$ hours on $\langle 80 \rangle$ CPU cores.

Stage 3 screens compositions by the witness criteria of Algorithm 2 – the scalar and map scans – reserving the heavier level-2 DPS relaxation for any flagged pair. Testing a single pair takes on the order of seconds even when multithreaded, so the full quadratic sweep over a pool of thousands of states is infeasible; we therefore scan as large a slice as the budget allows. The portion on record covers $\langle 5,000 \rangle$ ordered pairs, with $\langle \text{zero} \rangle$ detections. This pairwise sweep is necessarily partial, which is exactly why the witness-restricted SDP probe of the next section – complete over the entire witness library – carries the weight of the argument.

No composition was found entangled in any test: every composite of two PPT maps screened as PPT and separable to within tolerance, leaving the PPT2 conjecture without a counterexample.

5.2 The witness-restricted SDP probe

The broad search asks whether any composition in a finite pool is entangled. A sharper question is whether the witnesses we hold can detect entanglement on the composition manifold *at all*, even when the state is chosen to make them fire as hard as possible. The two witness-restricted SDPs of Sections 4 Bound entangled states from a witness and 4 Direct PPT2 search by see-saw answer it.

The first, `min_ppt_witness` ((32)), minimises $\text{tr}[W\rho]$ over the entire PPT cone. A negative optimum exhibits a PPT entangled state that W detects. Over the library, `<all 10,000>` witnesses attained a negative optimum, with margins ranging from about `<10-7 to <3 × 10-2 (median <5 × 10-5). This confirms that the canonical witnesses do carry genuine PPT-detection power once the state is optimised to them, as Theorem 3.15 predicts, in contrast to their inertness on randomly drawn states (section 5 Representation and symmetry effects); the small typical margin is itself the near-boundary effect.`

The second, `min_ppt2_witness` ((33)), restricts that minimisation from the whole PPT cone down to the composition manifold via the see-saw of section 4 Direct PPT2 search by see-saw, here run from `<32>` random restarts of at most `<64>` steps each.

The contrast between the two is the central result of this chapter. Every one of the 10,000 witnesses is live over the full PPT cone; yet when we ran the heavier see-saw on a sample of `<2,000>` of them, `<not one>` detected anything on the composition manifold. Every see-saw optimum was non-negative, lying in `<[2 × 10-16, 2 × 10-9]>` — flush against the separable boundary rather than crossing it. The witnesses that are demonstrably *live* over the PPT cone go *dead* once the state is constrained to be a composition of two PPT maps, exactly the signature expected if PPT2 holds in 4×4 : the composition manifold lies within the entanglement-breaking region, beyond the reach of every witness we can construct.

5.3 Detection methods compared

Recording every criterion's score on each state lets us compare the two detection routes directly, the comparison made meaningful by the DPS–PNCP correspondence. In practice they are far from equivalent. Of the `<849>` random bound entangled states that the level-2 DPS relaxation certifies (section 5 Search scale and the broad composition search), the PNCP witness library detects `<none>` by the scalar test: all `<8.5 million>` witness-state evaluations were strictly positive, the smallest being `<+148>`. The tight, near-decomposable KMSZ witnesses thus entirely miss the generic bound entangled states that the DPS relaxation reaches, the inertness of section 5 Representation and symmetry effects made quantitative.

Within the witness scan itself, the map witness test is the stronger of its two forms: the inequality $\lambda_{\min}((I \otimes \Phi^*)(\rho)) \leq \text{tr}[C_{\Phi}\rho]$ of section 3 Testing entanglement guarantees it detects everything the scalar test does, and in

general strictly more, since it inspects the whole spectrum of $(I \otimes \Phi_W)(C)$ rather than a single expectation.

5.4 Witness generality: cross-detection

Each witness is built tight to one prescribed family of zeros, so it is natural to ask how far it reaches beyond the single state it was made for. We cross-evaluated every witness against the bound entangled state that the witness-restricted SDP extracted for every *other* witness, separately for the scalar criterion (`cross_trace.jl`) and the map criterion (`cross_ampl.jl`). The result is stark: across all $\langle 10^8 \rangle$ witness–state pairs, neither criterion produced a single foreign detection. Both tests flagged only the $\langle 10,000 \rangle$ diagonal pairs, each witness detecting exactly the one state it generated and $\langle \text{no} \rangle$ other. This vanishing cross-detection rate is the sharpest expression of the tightness of the witnesses: each cuts the entangled region in just one place, at its own prescribed state, so the library behaves as $\langle 10,000 \rangle$ essentially disjoint detectors rather than a redundant cover.

5.5 Representation and symmetry effects

Two further observations concern the witness representations themselves rather than the search outcome.

First, the partial-transpose-invariant (symmetric) representatives are inert for our purpose. Symmetrized witnesses, the $M_0 = M_0^\Gamma$ representatives returned by the construction, never flagged a single PPT entangled state in our experiments, and PPT states built in the symmetric `ppt_invariant` shape never appeared as counterexamples. This is *not* because they are decomposable: the canonical KMSZ witness is exactly this partial-transpose-invariant representative (section 3 Real maps and their complexification), and it is provably indecomposable (Theorem Theorem 3.15), so it does carry genuine PPT-detection power. The inertness is instead a near-boundary effect. As argued previously, these witnesses are only tight and sit close to the decomposable cone, so whatever PPT entanglement they detect does so with a margin at the level of solver noise – consistent with the small PPT-violation margins (of order 10^{-6}) reported for the same construction by [22]. In practice such detections fall below our tolerance, so a fixed canonical witness rarely fires on a randomly drawn bound entangled state. The robust detection in the all-pairs search therefore rests on the DPS relaxation; the witness library’s genuine reach is exposed instead when the state is optimised to the witness, which is the point of the witness-restricted SDP probe (section 5 The witness-restricted SDP probe). For the cheap per-state scan we use the canonical representative on generic, non-symmetrized candidates.

Second, we also probed the asymmetric representatives directly (section 4 A note on the representation freedom). On the real domain they behave exactly as the theory predicts: they remain positive on all real product vectors, and over many random trials never assigned a negative score to a real separable state. Whether this extends to the complex domain is open: the relations underlying the representation freedom (section 3 Non-uniqueness of the matrix representation) vanish only on real product vectors, so an asymmetric representative that is a valid witness over $\mathbb{R}$ need not be block-positive over $\mathbb{C}$. Without an analytic guarantee the asymmetric witnesses proved too unreliable to use: the map test in particular can return a negative eigenvalue on a separable input – a spurious detection – precisely because an asymmetric representative need not be a positive map over $\mathbb{C}$. Establishing complex block-positivity, or restoring it by a suitable projection, is left for future work (section 6 Future work).

A more direct comparison may also be with the recent paper [22], where the authors produce a library of 20000 KMSZ witnesses in 3×3 , stating it took “several days”, albeit on a marginally weaker machine. Our result is still orders of magnitude faster.

5.6 Performance

A direct head-to-head comparison with the earlier MATLAB prototype is difficult, because the two implementations were built with different goals and make different design choices. The prototype drew random *integer* matrices to produce cosmetically nicer certificates; this is not random enough and frequently failed, so a single candidate often needed many recomputation attempts. Our implementation can still generate candidates that way, but for speed it samples from a normal distribution, which almost always succeeds on the first try. The two also rationalize differently: the prototype rationalizes the entire problem up front, paying for rational arithmetic throughout, whereas we rationalize only after the SDP is solved (4 Generating a witness library). The solver itself is largely language-independent, but the Julia code benefits from more efficient data handling and tighter solver integration.

With those caveats, we report rough wall-clock figures for the current implementation. Generating both pools for the 4×4 search – the library of 10,000 PNCP witness maps and the bound entangled candidate pool – took approximately $\langle 5 \rangle$ hours on $\langle 80 \rangle$ CPU cores, i.e. on the order of $\langle ?? \text{ s per PNCP map} - \text{need per-map timing} \rangle$ per map amortised. For the same dimensions the prototype took orders of magnitude longer to produce candidates.

fill the MAT-LAB-prototype figure (measured or from the article) and the per-map Julia tim-

6 Discussion

This thesis develops a computational approach to the PPT2 conjecture and applies it to the smallest open case, 4×4 . The pipeline generates PPT candidate states, builds a library of provably indecomposable PNCP witnesses from the KMSZ construction, and screens composite channels for entanglement by two distinct routes – entanglement witnesses, and the level-2 DPS hierarchy – with a post-solver rationalization step that turns any numerical detection into an exact certificate.

No composition was ever found to be entangled. The strongest evidence comes not from the brute pairwise sweep, which the per-pair cost keeps necessarily partial, but from the witness-restricted SDP probe (section 5 The witness-restricted SDP probe): although every one of our witnesses detects a bound entangled state somewhere in the full PPT cone, not one detects anything once the state is constrained to be a composition of two PPT maps, with the see-saw optima sitting exactly on the separable boundary. Witnesses that are demonstrably live over the PPT cone go uniformly dead on the composition manifold, precisely the signature expected if the PPT2 conjecture holds.

This is computational evidence, not a proof, and it inherits the limitations of its tools. The KMSZ witnesses, though provably indecomposable (Theorem Theorem 3.15), are only tight and lie close to the decomposable cone, so they detect PPT entanglement weakly. Much of the detection burden fell on the DPS relaxation, itself confined to level 2 in this dimension. A genuine counterexample, should one exist, might also lie in a region that our random candidate generation never reaches. What the methodology does provide is a reproducible, exactly-validated framework that returns a clean negative across every test we could run, and that extends in principle to the next open dimensions.

6.1 Complexity and Practical Limits

All the problems we are looking at are SDPs with exponential [7] complexity in dimension and relaxation depth [17]. Practical bottlenecks include:

1. processing power: these problems are by nature not easily parallelizable, simplex methods are intrinsically sequential and internal point methods rely on factoring large *sparse* matrices, so the potential of GPU speedups is fairly low [15].

2. memory growth: the size of the SDP grows exponentially with dimension and relaxation depth, leading to memory bottlenecks even for moderate dimensions, i.e. for 4×4 states DPS level 3 is already infeasible on standard hardware, requiring hundreds of gigabytes of RAM. Even improvements such as adding KKT constraints to achieve finite convergence [18] lead to significant increases in problem size, so they are not without cost.
3. solver instability near degeneracy: SDP solvers produce floating-point solutions, and when the problem is near the boundary of feasibility, small numerical errors can lead to incorrect conclusions about positivity. This is particularly problematic for our purposes, since we are interested in the very existence of such bound states.
4. The generated witnesses are provably indecomposable (Theorem Theorem 3.15), so each one can in principle detect some PPT entangled state. They are, however, only tight and therefore lie close to the decomposable cone, detecting PPT entanglement only weakly. This compounds the numerical fragility of point 3, and how best to improve a given witness, remains open.
5. Searching for PPT candidates in the first place is non-trivial, and random generation may not be sufficient to find counterexamples if they exist. We may require a more structured approach if the volume of the search space is in fact 0 (this would not mean the conjecture holds).

6.2 Future work

Several directions remain open.

The most immediate is the *complex-domain extension of the representation freedom* (Sections 3 Non-uniqueness of the matrix representation, 4 A note on the representation freedom). The asymmetric Gram representatives are valid witnesses over $\mathbb{R}$ but lose block-positivity over $\mathbb{C}$, because the Segre relations spanning L vanish only on real product vectors. Klep et al. restore positivity over $\mathbb{C}$ for the canonical representative by annihilating the skew-symmetric part [20]; an analogous analytic guarantee for the asymmetric family, or a projection that restores complex block-positivity while keeping its sharper cut of the entangled region, would turn the freedom from a real-only curiosity into a usable supply of independent witnesses. We pursued this only numerically and found the evidence too weak to rely on, so the analytic question is left open.

A second direction is *structured candidate generation*. Random PPT sampling discards most of its draws as separable and leaves unstructured

survivors (limitation 5, section 6 Complexity and Practical Limits); if the volume of bound entangled compositions is in fact vanishing, a random search will never reach a counterexample even if one exists. The constructions surveyed in section Algorithm 1 – unextendible product bases [27], antisymmetric-subspace states [28], and symmetric random induced states [29] – offer entanglement-by-design pools that a targeted search could exploit.

Finally, the pipeline is dimension-limited by the cost of the SDPs (section 6 Complexity and Practical Limits). Pushing beyond 4×4 , whether by exploiting state symmetry [19], by the finite-convergence KKT extensions of [18], or by specialised solvers, would let the same methodology test the conjecture where even less is known.

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